

Definition The **dot (inner or scalar) product** of two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is the scalar given by

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

Example. Find $\mathbf{u} \cdot \mathbf{v}$ when $\mathbf{u} = \langle 1, -2, -1 \rangle$ and $\mathbf{v} = \langle -6, 2, -3 \rangle$

Theorem. If θ is the angle between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, then

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \quad \text{or} \quad \theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right).$$

Example. Find the angle between $\mathbf{u} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$.

Theorem. Let θ be the angle between two vectors $\mathbf{u}$ and $\mathbf{v}$.

(i) θ is acute when $\mathbf{u} \cdot \mathbf{v} > 0$ and obtuse when $\mathbf{u} \cdot \mathbf{v} < 0$

(ii) $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular) if and only if $\mathbf{u} \cdot \mathbf{v} = 0$

Example. Determine whether the vectors $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + 1\mathbf{k}$ and $\mathbf{v} = 2\mathbf{j} + 4\mathbf{k}$ are orthogonal.

Theorem. Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be three vectors and let c be a scalar.

- (i) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- (ii) $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
- (iii) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
- (iv) $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
- (v) $\vec{0} \cdot \mathbf{u} = 0$

We now take two vectors $\mathbf{u} = \overrightarrow{PQ}$ and $\mathbf{v} = \overrightarrow{PS}$ and ask “How much of $\mathbf{u}$ can be found in $\mathbf{v}$?” The answer turns out to be $\mathbf{u}$ ’s ‘shadow’ on $\mathbf{v}$:

This ‘shadow’ is known as a vector projection. Specifically, it is the **vector projection** of $\mathbf{u}$ onto $\mathbf{v}$. It is determined by ‘dropping a perpendicular’ from the point Q onto the line PS . It is notated by $\text{proj}_{\mathbf{v}}(\mathbf{u})$. We compute it in the following manner:

$$\begin{aligned}
 \text{proj}_{\mathbf{v}}(\mathbf{u}) &= (\text{distance traveled})(\text{directional unit vector}) \\
 &= (|\mathbf{u}| \cos(\theta)) \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right) \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}.
 \end{aligned}$$

Note: $\mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|}$ is called the *scalar projection* or *scalar component* of $\mathbf{u}$ in the direction of $\mathbf{v}$.

Example. Given the vectors $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ determine $\text{proj}_{\mathbf{v}}(\mathbf{u})$.

Example. Given the vectors $\mathbf{v} = \langle 6, 3, -2 \rangle$ and $\mathbf{u} = \langle 3, 4, 0 \rangle$ determine $\text{proj}_{\mathbf{v}}(\mathbf{u})$.